

OS ASSIGNMENT 1 (24K-1033)

Hello World syscall

```
kernel - Thunar
Terminal - root@unperturbable-se

Terminal - root@unperturbable-se: /home/unperturbable-se/Desktop/kernel/linux 6.19.3

/*
#include <linux/kernel.h>

asmlinkage long sys_helloworld(void)
{
    printk("Hello World!!!!\n I AM K241033 YYYYYYYY!!! WOOHOOO!\n");
    return 0;
}

#include <linux/kernel.h>
#include <linux/syscalls.h>

SYSCALL_DEFINE0(helloworld)
{
    printk("Hello World!!!!\n I AM K241033 YYYYYYYY!!! WOOHOOO! CHIMPANZINI BANANINI WAWAWA.....TUNG TUNG TUNG TUNG \n");
    return 0;
}
```

```
Terminal - unperturbable-se
Terminal - unperturbable-se@unperturbable-se-VirtualBox: ~/Desktop/kernel/linux 6.19.3

GNU nano 7.2 arch/x86/entry/syscalls/syscall_64.tbl
383 common name to handle at sys name to handle at
384 common open by handle at sys open by handle at
385 common clock_adjtime sys clock_adjtime
386 common syncfs sys syncfs
387 64 sendmsg sys sendmsg
388 common setns sys setns
389 common getcpu sys getcpu
310 64 process_vm_readv sys process_vm_readv
311 64 process_vm_writev sys process_vm_writev
312 common kcmp sys kcmp
313 common finit_module sys finit_module
314 common sched_setattr sys sched_setattr
315 common sched_getattr sys sched_getattr
316 common renameat2 sys renameat2
317 common seccomp sys seccomp
318 common getrandom sys getrandom
319 common memfd_create sys memfd_create
320 common kexec_file_load sys kexec_file_load
321 common bpf sys bpf
322 64 execveat sys execveat
323 common userfaultfd sys userfaultfd
324 common membarrier sys membarrier
325 common mlock2 sys mlock2
326 common copy_file_range sys copy_file_range
327 64 preadv2 sys preadv2
328 64 pwritev2 sys pwritev2
329 common pkey_mprotect sys pkey_mprotect
330 common pkey_alloc sys pkey_alloc
331 common pkey_free sys pkey_free
332 common statx sys statx
333 common io_pgetevents sys io_pgetevents
334 common rseq sys rseq
335 common uretprobe sys uretprobe
336 common uprobe sys uprobe
337 64 helloworld sys helloworld
# don't use numbers 387 through 423, add new calls after the last
# 'common' entry
424 common pidfd_send_signal sys pidfd_send_signal
425 common io_uring_setup sys io_uring_setup
426 common io_uring_enter sys io_uring_enter
427 common io_uring_register sys io_uring_register
428 common open_tree sys open_tree
429 common move_mount sys move_mount
430 common fsopen sys fsopen
431 common fsconfig sys fsconfig
432 common fsmount sys fsmount
433 common fspick sys fspick
434 common pidfd_open sys pidfd_open

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward Prev Word Next Word Home End
```

```
Terminal - unperturbable-se
Terminal - unperturbable-se@unperturbable-se-VirtualBox: ~/Desktop/kernel/linux-6.19.3

GNU nano 7.2 include/linux/syscalls.h
/* obsolete */
asmlinkage long sys_gethostname(char __user *name, int len);
asmlinkage long sys_uname(struct old_utsname __user *);
asmlinkage long sys_olduname(struct oldold_utsname __user *);
#ifdef __ARCH_MANT_SYS_OLD_GETRLIMIT
asmlinkage long sys_old_getrlimit(unsigned int resource, struct rlimit __user *rlim);
#endif

/* obsolete */
asmlinkage long sys_ipc(unsigned int call, int first, unsigned long second,
                        unsigned long third, void __user *ptr, long fifth);

/* obsolete */
asmlinkage long sys_mmap_pgoff(unsigned long addr, unsigned long len,
                              unsigned long prot, unsigned long flags,
                              unsigned long fd, unsigned long pgoff);
asmlinkage long sys_old_mmap(struct mmap_arg_struct __user *arg);

/*
 * Not a real system call, but a placeholder for syscalls which are
 * not implemented -- see kernel/sys.n.c
 */
asmlinkage long sys_ni_syscall(void);
asmlinkage long sys_helloworld(void);
#endif /* CONFIG_ARCH_HAS_SYSCALL_WRAPPER */

asmlinkage long sys_ni_posix_timers(void);

/*
 * Kernel code should not call syscalls (i.e., sys_xyzzy()) directly.
 * Instead, use one of the functions which work equivalently, such as
 * the ksys_xyzzy() functions prototyped below.
 */
ssize_t ksys_write(unsigned int fd, const char __user *buf, size_t count);
int ksys_fchown(unsigned int fd, uid_t user, gid_t group);
size_t ksys_read(unsigned int fd, char __user *buf, size_t count);
void ksys_sync(void);
int ksys_unshare(unsigned long unshare_flags);
int ksys_setsid(void);
int ksys_sync_file_range(int fd, loff_t offset, loff_t nbytes,
                        unsigned int flags);
ssize_t ksys_pread64(unsigned int fd, char __user *buf, size_t count,
                    loff_t pos);
size_t ksys_pwrite64(unsigned int fd, const char __user *buf,
                    size_t count, loff_t pos);
int ksys_fallocate(int fd, int mode, loff_t offset, loff_t len);
#endif CONFIG_ADVISE_SYSCALLS

Help Write Out Where Is Cut Execute Location Undo Set Mark To Bracket Previous Back Prev Word Next Word Home End
Exit Read File Replace Paste Justify Go To Line Redo Copy Where Was Next Forward Next Word End
```

```
Terminal - unperturbable-se
Terminal - unperturbable-se@unperturbable-se-VirtualBox: ~/Desktop/kernel/linux-6.19.3

GNU nano 7.2 kernel/Makefile
# SPDX-License-Identifier: GPL-2.0
#
# Makefile for the Linux kernel.
#
obj-y = fork.o exec_domain.o panic.o helloworld.o \
       cpuclock.o softirq.o resource.o \
       sysctl.o capability.o ptrace.o user.o \
       signal.o sys.o umh.o workqueue.o pid.o task_work.o \
       extable.o param.o \
       kthread.o sys_mn.o nsproxy.o nstree.o nscommon.o \
       notifier.o ksyzfs.o cred.o reboot.o \
       async.o range.o smptboot.o ucount.o reset.o ksyms_common.o

obj-$(CONFIG_MULTITIMER) += groups.o
obj-$(CONFIG_VHOST_TASK) += vhost_task.o

ifdef CONFIG_FUNCTION_TRACER
# Do not trace internal ftrace files
CFLAGS_REMOVE_irq_work.o = -x $(CFLAGS_FTRACE)
endif

# Branch profiling isn't maintr-safe
ifdef CONFIG_TRACE_BRANCH_PROFILING
CFLAGS_context_tracking.o += -DNOTABLE_BRANCH_PROFILING
endif

# Prevents flicker of uninteresting __do_softirq()/__local_bh_disable_ip()
# in coverage traces.
KCOV_INSTRUMENT_softirq.o := n
# Avoid KCSAN instrumentation in softirq ("No shared variables, all the data
# are CPU local" => assume no data races), to reduce overhead in interrupts.
KCSAN_SANITIZE_softirq.o := n
# These are called from save_stack_trace() on slub debug path,
# and produce insane amounts of uninteresting coverage.
KCOV_INSTRUMENT_extable.o := n
KCOV_INSTRUMENT_stacktrace.o := n
# Don't self-instrument.
KCOV_INSTRUMENT_kcov.o := n
# If sanitizers detect any issues in kcov, it may lead to recursion
# via printk, etc.
KASAN_SANITIZE_kcov.o := n
KCSAN_SANITIZE_kcov.o := n
UBSAN_SANITIZE_kcov.o := n
KASAN_SANITIZE_kcov.o := n
CFLAGS_kcov.o += $(call cc-option, -fno-conserve-stack) -fno-stack-protector

obj-y += sched/
```

Kernel Compilation


```
Terminal - root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3
root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3
cc arch/x86/boot/compressed/cputags.o
cc arch/x86/boot/compressed/early_serial_console.o
cc arch/x86/boot/compressed/asmL.o
cc arch/x86/boot/compressed/ident_map_64.o
cc arch/x86/boot/compressed/ldt_64.o
AS arch/x86/boot/compressed/ldt_handlers_64.o
AS arch/x86/boot/compressed/mem_encrypt.o
cc arch/x86/boot/compressed/pgtable_64.o
cc arch/x86/boot/compressed/sev.o
cc arch/x86/boot/compressed/sev-handle-vc.o
cc arch/x86/boot/compressed/acpi.o
cc arch/x86/boot/compressed/tdx.o
AS arch/x86/boot/compressed/tdxcall.o
cc arch/x86/boot/compressed/tdx-shared.o
cc arch/x86/boot/compressed/mem.o
cc arch/x86/boot/compressed/efi.o
LD arch/x86/boot/compressed/vmlinux
ZOFFSET arch/x86/boot/zoffset.h
AS arch/x86/boot/header.o
cc arch/x86/boot/main.o
cc arch/x86/boot/memory.o
cc arch/x86/boot/pm.o
AS arch/x86/boot/pmlump.o
cc arch/x86/boot/printk.o
cc arch/x86/boot/regs.o
cc arch/x86/boot/string.o
cc arch/x86/boot/tty.o
cc arch/x86/boot/video.o
cc arch/x86/boot/video-mode.o
cc arch/x86/boot/version.o
cc arch/x86/boot/video-vga.o
cc arch/x86/boot/video-vesa.o
cc arch/x86/boot/video-bios.o
LD arch/x86/boot/setup.elf
OBJCOPY arch/x86/boot/setup.bin
OBJCOPY arch/x86/boot/vmlinux.bin
BUILD arch/x86/boot/bzImage
kernel: arch/x86/boot/bzImage is ready (#0)
root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3# ls -lh arch/x86/boot/bzImage
-rw-r--r-- 1 root root 14M Feb 28 16:14 arch/x86/boot/bzImage
root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3# file arch/x86/boot/bzImage
arch/x86/boot/bzImage: Linux kernel x86 boot executable bzImage, version 6.24.1033 (root@unperturbable-se-VirtualBox) #6 SMP PREEMPT_DYNAMIC Sat Feb 28 16:10:47 PKT 2026, RO-rootfs, swap_dev 0x0, Normal VGA
root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3# # This command checks the compressed data integrity
gzip -t < arch/x86/boot/bzImage
gzip: stdin: not in gzip format
root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3# grep "sys_helloworld" System.map
ffffffffff136db70 t pfx do_sys_helloworld
ffffffffff136db70 t pfx ia32_sys_helloworld
ffffffffff136db70 t pfx x64_sys_helloworld
ffffffffff136db80 t do_sys_helloworld
ffffffffff136db80 t ia32_sys_helloworld
```

Creating microkernel modules

```
Terminal - root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3
root@unperturbable-se-VirtualBox: /home/unperturbable-se/Desktop/kernel/linux-6.19.3# sudo make modules_install
SYMLINK /lib/modules/6.24.1033/build
INSTALL /lib/modules/6.24.1033/modules.order
INSTALL /lib/modules/6.24.1033/modules.builtin
INSTALL /lib/modules/6.24.1033/modules.builtin.modinfo
INSTALL /lib/modules/6.24.1033/modules.builtin.ranges
INSTALL /lib/modules/6.24.1033/kernel/arch/x86/events/rapl.ko
SIGN /lib/modules/6.24.1033/kernel/arch/x86/events/rapl.ko
INSTALL /lib/modules/6.24.1033/kernel/arch/x86/kernel/msr.ko
SIGN /lib/modules/6.24.1033/kernel/arch/x86/kernel/msr.ko
INSTALL /lib/modules/6.24.1033/kernel/arch/x86/crypto/aesni-intel.ko
SIGN /lib/modules/6.24.1033/kernel/arch/x86/crypto/aesni-intel.ko
INSTALL /lib/modules/6.24.1033/kernel/arch/x86/crypto/ghash-clmulni-intel.ko
SIGN /lib/modules/6.24.1033/kernel/arch/x86/crypto/ghash-clmulni-intel.ko
INSTALL /lib/modules/6.24.1033/kernel/kernel/configs.ko
SIGN /lib/modules/6.24.1033/kernel/kernel/configs.ko
INSTALL /lib/modules/6.24.1033/kernel/fs/bifmt_misc.ko
SIGN /lib/modules/6.24.1033/kernel/fs/bifmt_misc.ko
INSTALL /lib/modules/6.24.1033/kernel/fs/autofs/autofs4.ko
SIGN /lib/modules/6.24.1033/kernel/fs/autofs/autofs4.ko
INSTALL /lib/modules/6.24.1033/kernel/fs/vboxsf/vboxsf.ko
SIGN /lib/modules/6.24.1033/kernel/fs/vboxsf/vboxsf.ko
INSTALL /lib/modules/6.24.1033/kernel/crypto/hkdf.ko
SIGN /lib/modules/6.24.1033/kernel/crypto/hkdf.ko
INSTALL /lib/modules/6.24.1033/kernel/crypto/cryptd.ko
SIGN /lib/modules/6.24.1033/kernel/crypto/cryptd.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/video/fbdev/vga16fb.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/video/fbdev/vga16fb.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/video/vgastate.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/video/vgastate.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/video/vgastate.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/acpi/video.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/acpi/video.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/char/lp.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/char/lp.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/char/ppdev.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/char/ppdev.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/gpu/drm/ttm_helper.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/gpu/drm/ttm_helper.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/gpu/drm/ttm.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/gpu/drm/ttm.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/gpu/drm/vmwgfx/vmwgfx.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/gpu/drm/vmwgfx/vmwgfx.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/macintosh/mac_hid.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/macintosh/mac_hid.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/nvme/common/nvme-auth.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/nvme/common/nvme-auth.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/nvme/common/nvme-keyring.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/nvme/common/nvme-keyring.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/nvme/host/nvme-core.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/nvme/host/nvme-core.ko
INSTALL /lib/modules/6.24.1033/kernel/drivers/nvme/host/nvme.ko
SIGN /lib/modules/6.24.1033/kernel/drivers/nvme/host/nvme.ko
```

Success

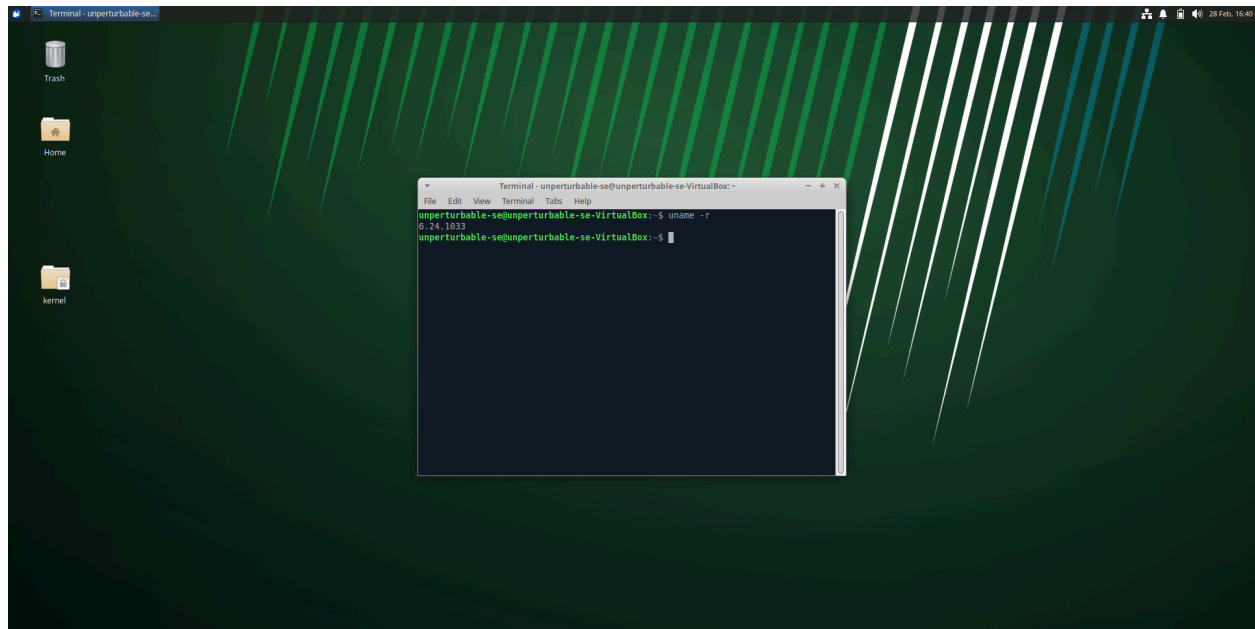
```
run-parts: executing /etc/kernel/postinst.d/xx-update-initrd-links 6.24.1033 /boot/vmlinuz-6.24.1033
I: /boot/initrd.img is now a symlink to initrd.img-6.24.1033
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 6.24.1033 /boot/vmlinuz-6.24.1033
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.24.1033
Found initrd image: /boot/initrd.img-6.24.1033
Found linux image: /boot/vmlinuz-6.17.0-14-generic
Found initrd image: /boot/initrd.img-6.17.0-14-generic
Found memtest86+x64 image: /boot/memtest86+x64.bin
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB DISABLE OS PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
```

Done(role number = 24.1033)

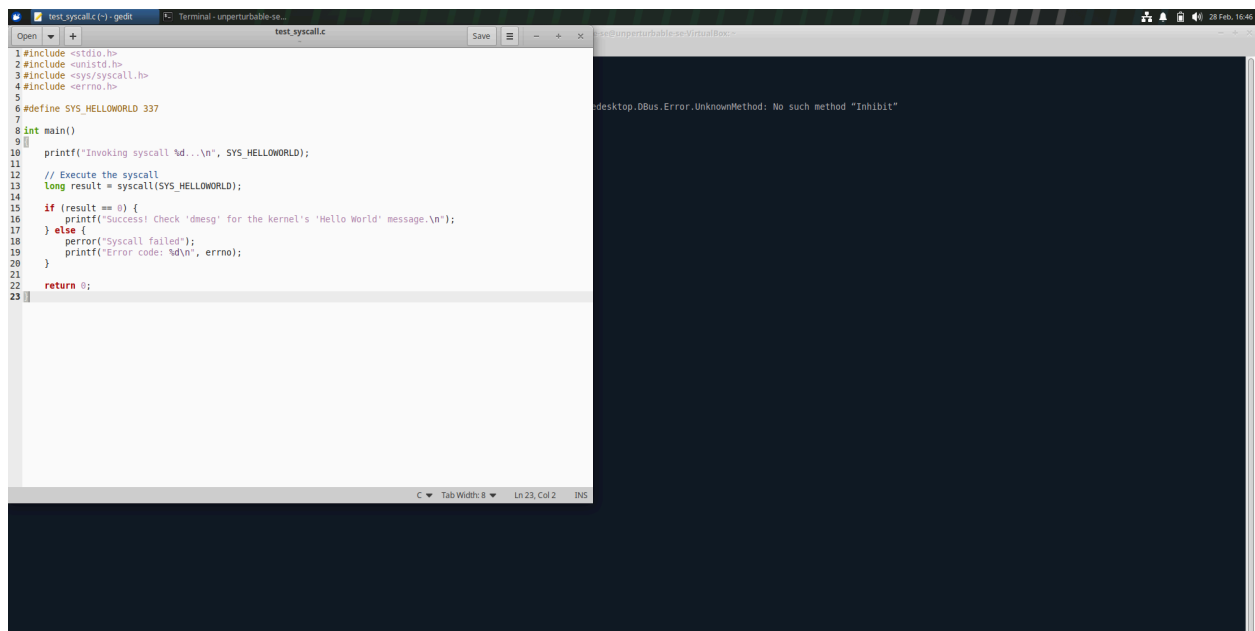
GNU GRUB version 2.12

```
*Ubuntu, with Linux 6.24.1033
Ubuntu, with Linux 6.24.1033 (recovery mode)
Ubuntu, with Linux 6.17.0-14-generic
Ubuntu, with Linux 6.17.0-14-generic (recovery mode)
```

Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands
before booting or 'c' for a command-line. ESC to return previous
menu.



Testing helloworld syscall



```
unperturbable-se@unperturbable-se-VirtualBox:~$ gcc test_syscall.c -o test_syscall
```

```
unperturbable-se@unperturbable-se-VirtualBox:~$ ./test_syscall
```

```
Invoking syscall 337...
```

```
Success! Check 'dmesg' for the kernel's 'Hello World' message.
```

```
unperturbable-se@unperturbable-se-VirtualBox:~$ sudo dmesg | tail -n 1
```

```
        I AM K241033 YAYYYYYY!!! WOOH000!      CHIMPANZINI BANANINI WAWAWA.....TUNG TUNG TUNG TUNG  
unperturbable-se@unperturbable-se-VirtualBox:~$ █
```